Example 4 - Solve 3x3 Systems of Equations using Gauss Elimination

Example:
Gauss - Elimination 3x3 system
2 x - 3 y + 3 z = 7
3 x + 1 y - 2 z = -11
5 x - 2 y + 4 z = 11

Solution:
nasty coefficients
make $a_{11} = 1$
2 x - 3 y + 3 z = 7 <---- (row 1) / 2 or R1 / 2
3 x + 1 y - 2 z = -11
5 x - 2 y + 4 z = 11

make $a_{21} = 0$ & $a_{31} = 0$
1 x - 1.5 y + 1.5 z = 3.5
3 x + 1 y - 2 z = -11 <---- (row 2) - 3 (row 1)
5 x - 2 y + 4 z = 11 <---- (row 3) - 5 (row 1) or R3 - 5 R1

make $a_{22} = 1$
1 x - 1.5 y + 1.5 z = 3.5
0 x + 5.5 y - 6.5 z = -21.5 <---- (row 2) / 5.5
0 x + 5.5 y - 3.5 z = -6.5

make $a_{32} = 0$
1 x - 1.5 y + 1.5 z = 3.5
0 x + 1 y - 1.18 z = -3.91
0 x + 5.5 y - 3.5 z = -6.5 <---- (row 3) - 5.5 (row 2)

make $a_{33} = 1$
1 x - 1.5 y + 1.5 z = 3.5
0 x + 1 y - 1.18 z = -3.91
0 x + 0 y + 3 z = 15 <---- (row 3) / 3

Now in Gauss Form
1 x - 1.5 y + 1.5 z = 3.5
0 x + 1 y - 1.18 z = -3.91
0 x + 0 y + 1 z = 5
"ones" on the diagonal and "zeroes" below the diagonal

... from row (3): z = 5
back sub. in row (2) --> y = -3.91 + 1.18 (5) = 2.0 ; y = 2.0

back sub. in row (1) --> x = (1.5) (2.0) - (1.5) (5) + 3.5 = -1 ; $\underline{x = -1}$

$\therefore$ answer (x , y , z) = (-1 , 2 , 5)